

SARANYA GANESAN

JAVA BACKEND ENGINEER | SPRING BOOT | MICROSERVICES | KAFKA | GRAPHQL

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PROFESSIONAL SUMMARY

Backend Engineer with 5+ years of experience building enterprise APIs and distributed systems across airline, financial-services, telecom and startup environments. Strong in **Java 21, Spring Boot 3.x, Microservices, Kafka, GraphQL/Apollo Federation, PostgreSQL, AWS and Kubernetes/OpenShift**. Proven ownership of greenfield services, stateless aggregation/service orchestration, event-driven architecture, API security, CI/CD quality gates, production observability and incident resolution.

TECHNICAL SKILLS

Backend	Java 21, Spring Boot 3.x, Spring MVC, WebClient, Spring Data JPA/Hibernate, Node.js, Express.js
Architecture & APIs	Microservices, REST, GraphQL, Apollo Federation/Router, Stateless Aggregation, Service Orchestration, Event-Driven Architecture
Messaging & Data	Kafka, Kafka Connect, JSON, Avro, PostgreSQL, MariaDB, MySQL, Liquibase, Snowflake
Cloud & Platform	AWS S3, Kubernetes, OpenShift, Docker, Helm, IBM Cloud/KPaaS, Vault, Apigee, Akamai, Istio
Security & Delivery	OAuth2, JWT, IBM Cloud Secrets Manager, GitHub Actions, Trident CI/CD, JaCoCo
Testing & Observability	JUnit, Mockito, WireMock, H2/Embedded PostgreSQL, Contract Testing, Apollo Studio, Dynatrace, Mezmo, Instana, Kibana

PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions - GraphQL / Platform API Developer

Client: American Airlines | Web Platform GraphQL - ReAccommodation | May 2026 - Present | Fort Worth, TX

- **Designed and built ReAccommodation (ReAccom) from scratch** using **Java 21, Spring Boot and Apollo Federation**; defined 3 queries + 1 mutation for IROP eligibility, reservation/disruption details, alternate-flight search and passenger rebook/reissue.
- **Engineered a stateless aggregation/service-orchestration layer** with no local DB, composing live reservation, check-in and flight data from distributed airline services through reactive Spring WebClient.
- **Configured downstream HTTP controls** including a **10-second response timeout, max 60 connections, 15-second pending-acquire timeout and connection lifecycle limits** for enterprise integrations.
- **Support observed production traffic of ~418 requests/min across two regions** (268 RPM + 150 RPM), using Apollo Studio, Mezmo and Dynatrace to investigate GraphQL operations, downstream failures and application errors.
- **Prevented recurrence of a federation-wide schema failure** by identifying a root-level authentication directive that affected composed subgraphs and contributed to **~2 days of blocked/disrupted composition activity**; implemented a pre-composition schema quality gate.
- **Strengthened CI engineering controls** with JaCoCo coverage, Java/Spring Boot standards and GraphQL schema validation; automated synchronization of centrally maintained rules into repository-level compliance checks.
- **Migrated 15+ application credentials** from GitHub-managed secrets to **IBM Cloud Secrets Manager** and support JWT/scope handling plus service-to-service OAuth2 client-credentials authentication through Apigee.
- **Resolved a high-impact production authentication incident within one business day** by tracing expired-token failures through Router/coprocessor flows and identifying concatenated stale/current credentials; participate in rotational xMatters/GSC resiliency support.
- **Contribute across the Web Platform ecosystem** through pair programming/support for Customer Data Graph, Session API, Aileron logging and Smart Links, spanning Java/Spring Boot, Node.js and Redis-backed platform capabilities.

Tech: Java 21, Spring Boot, WebClient, GraphQL, Apollo Federation/Router/Studio, REST/SOAP, OAuth2/JWT, Apigee, GitHub Actions, JaCoCo, Kubernetes/KPaaS, IBM Cloud Secrets Manager, Dynatrace, Mezmo

ONEngine AI - Backend Engineer

Jun 2025 - Mar 2026

- **Built backend capabilities for a work-order/property-restoration platform** covering quotations, approvals, workforce/material assignment, job execution and claims workflows; implemented Liquibase schema changes for evolving requirements.
- **Prototyped a document-management service** using **AWS S3 presigned URLs, Docker and Kubernetes** as part of architecture evaluation during a platform technology transition.
- **Adapted to the redesigned Node.js/Express.js/Prisma stack** and developed a **work-order-scoped collaboration module** for coordinators and field personnel.
- **Implemented membership-based chat access**, dynamic add/remove participants, persistent message history and attachment sharing; contributed Teams/Slack-oriented collaboration capabilities in development.

Tech: Node.js, Express.js, Prisma, REST APIs, Liquibase, AWS S3/Presigned URLs, Docker, Kubernetes; Laravel architecture POC

Career Break - Work Authorization Transition | Sep 2024 - May 2025

Capgemini America - Java Backend Engineer

Client: Discover Financial Services | Collections Value Stream - Payment Assistance / Decisioning | Jan 2023 - Aug 2024

- **Designed and built Collections Pre-D (coll-pre-d) from scratch** using **Java 21 and Spring Boot**, implementing REST APIs, PostgreSQL/JPA persistence, health checks, JWT authentication and enterprise integrations for payment-assistance decisioning.
- **Implemented relational/data workflows** with PostgreSQL, JPA/Hibernate, native queries and Liquibase; built an **AWS S3 CSV ingestion flow** to transform source records into application entities for persistence.
- **Developed Kafka producer/consumer workflows** with **@RetryableTopic backoff, dead-letter topics, idempotent producers, transactions, offset management and consumer groups** using JSON/Avro payloads.
- **Worked with Kafka Connect source and sink connectors** for Kafka/external-system data movement as part of the collections integration architecture.
- **Maintained 80%+ JUnit coverage** and built integration/acceptance tests with WireMock, H2 and embedded PostgreSQL; created account-key-driven service virtualization plus consumer-contract and container smoke tests.
- **Upgraded backend APIs from Spring Boot 2.x to 3.1.2 on Java 21**, migrating javax.* to Jakarta and resolving framework dependency/test compatibility issues.
- **Contributed to PCF-to-OpenShift modernization** by refactoring Trident CI/CD and Helm deployment configuration, onboarding Vault and integrating smoke tests; supported Multi-AZ and blue-green readiness.
- **Configured Instana service-reliability alerts** for downstream connectivity failures and supported JWT v3, Kibana/Instana observability and production go-live readiness for Collection Account Status.

Tech: Java 21, Spring Boot 3.1.2, JPA/Hibernate, REST, PostgreSQL, Liquibase, Kafka/Kafka Connect, JSON/Avro, AWS S3, OpenShift, Helm, Vault, JWT, JUnit, WireMock, Instana, Kibana

Infosys - Senior Systems Engineer

Client: Verizon | Oct 2022 - Dec 2022 | Remote

- **Completed advanced Spring Boot microservices/distributed-backend training** covering service decomposition, REST API design, logging and production engineering practices.

Career Break - Relocation & Family Care | Jan 2022 - Sep 2022

Collabera - Java Backend Developer

Client: Verizon | Jun 2021 - Dec 2021 | Remote

- **Resolved production defects across Java backend services and REST APIs**; performed root-cause analysis using logs, SQL/database queries and debugging tools in an SLA-driven environment.

Career Break - Family Care | Jun 2018 - May 2021

IVTL Infoview - Software Engineer

Enterprise Application Modernization | Jun 2016 - May 2018 | Chennai, India

- **Developed 20+ dynamic web interfaces** with HTML, CSS, JavaScript and AJAX and implemented Java backend logic integrated with PostgreSQL for legacy-to-web modernization.

PROJECT

LifePilot - AI Multi-Agent Personal Planner

- **Built a multi-agent planning system** using Google Gemini AI, embeddings/vector memory and automated task routing.
- **Developed a Streamlit UI and containerized deployment** using Docker and Google Cloud Run.

EDUCATION & CERTIFICATIONS

B.Tech - Information Technology | Sona College of Technology | 2012-2016 | GPA: 8.15/10

Certifications: OCI 2025 Foundations Associate | SAFe 5 Practitioner | Cambridge Business English | Kaggle AI Agents Intensive (2025)